Spectral Proper Orthogonal Decomposition of Wind-over-Waves DNS Data

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Abstract. This report presents a Spectral Proper Orthogonal Decomposition (SPOD) analysis of a Direct Numerical Simulation dataset of a wind-over-waves boundary layer. The decomposition is performed on a $192 \times 192 \times 65$ grid using 500 temporal snapshots, yielding 26 resolved frequencies from a block length of 50 snapshots and $N_{\text{blk}} = 19$ overlapping Welch blocks for ensemble averaging. The weighted Method of Snapshots is implemented with a non-uniform spatial weight matrix to compensate for the wall-normal grid clustering. The analysis identifies a sharp global energy peak at $f = 2.67$, where the leading mode captures 33.9% of the local spectral energy and corresponds to a streamwise-dominated coherent structure ($U : V : W = 68 : 21.6 : 10$). A coherent–incoherent transition appears at $f \approx 18.67$ from the mode-1 coherence ratio. At $f = 16.0$, the leading mode collapses onto a thin surface-locked layer; the wall-normal component reaches 35.5%, which is the only frequency where W exceeds V. The streamwise share drops with frequency: from a U-dominated coherent regime to a near-isotropic incoherent plateau ($U : V : W \approx 27 : 40 : 33$), with U crossing equipartition at $f = 26.67$. The phase-energy diagnostic shows all examined modes are travelling downstream rather than standing in place. Source code: github.com/sudinbhujushrestha/Spectral-POD.

1 Introduction

1.1 Motivation

Standard Proper Orthogonal Decomposition (POD) ranks the spatial structures of a flow by energy, but it has no notion of time. If two physical mechanisms live at different frequencies but happen to share a region of space, POD will fold them into the same mode. Spectral POD (SPOD) avoids this by keeping the time information [2, 1]: a temporal Fourier transform is applied to the space–time correlation tensor, giving the Cross-Spectral Density (CSD) tensor $\hat{C}(\boldsymbol{x}, \boldsymbol{x}', f)$. Diagonalising this at each frequency produces modes that are coherent in space and time. Vortex shedding or wave-coherent boundary-layer structures, for example, end up isolated at the frequency where they actually live.

This report applies SPOD to a Direct Numerical Simulation (DNS) dataset of a wind-over-waves boundary layer. The grid contains $N_x \times N_y \times N_z = 192 \times 192 \times 65$ points, and the snapshot ensemble comprises 500 temporal samples covering the streamwise, spanwise, and wall-normal velocity components (U, V, W). The questions of interest here are: which coherent structures are dominant, how are they distributed in the wall-normal direction, and where in the frequency band does the flow stop being coherent?

1.2 Welch's Method for Cross-Spectral Density Estimation

Estimating the CSD properly needs an ensemble of independent realisations, but the DNS gives only one long time series. Welch's method gets around this by chopping the record into N_{blk} overlapping blocks and Fourier-transforming each one [4, 1]. The block length controls a familiar trade-off: long blocks resolve closely-spaced frequencies but produce few realisations, short blocks do the opposite. For a selected frequency f_j, the Fourier coefficients across all blocks are gathered into the frequency-domain snapshot matrix

$$X(f_j) = \left[\hat{\boldsymbol{q}}^{(1)}(f_j),\, \hat{\boldsymbol{q}}^{(2)}(f_j),\, \ldots,\, \hat{\boldsymbol{q}}^{(N_{\text{blk}})}(f_j)\right] \in \mathbb{C}^{3N \times N_{\text{blk}}}.$$

1.3 The Spatial Weight Matrix

The weight matrix W sets the inner product that makes the spatial modes optimal and orthogonal. For an incompressible flow it is just a quadrature rule: each grid point gets weighted by the volume it controls. This correction is essential whenever the computational mesh is non-uniform: in the present DNS, the wall-normal direction is clustered near the surface to resolve the boundary-layer shear, so that without spatial weighting the densely-packed near-surface nodes would dominate the kinetic-energy calculation through grid density alone, regardless of the actual flow energy at those points. The construction of W for the present grid is detailed in Section 4.2.

[1] Source code available at: github.com/sudinbhujushrestha/Spectral-POD

2 Theoretical Formulation

This section derives the weighted Method of Snapshots used in the pipeline. The point is to see exactly where the grid weight matrix W enters both in the reduced eigenvalue problem and in the final spatial mode reconstruction.

2.1 Weighted Method of Snapshots

The theoretical weighted SPOD eigenvalue problem at a specific frequency f_j is

$$\hat{C}W\hat{\Phi} = \hat{\Phi}\hat{\Lambda}. \tag{1}$$

Substituting the empirical CSD estimate $\hat{C} = \frac{1}{N_{\text{blk}}}XX^*$ yields

$$\frac{1}{N_{\text{blk}}}XX^*W\hat{\Phi} = \hat{\Phi}\hat{\Lambda}. \tag{2}$$

Because XX^*W is an asymmetric matrix of size $3N \times 3N$, this system cannot be solved efficiently by direct decomposition. Multiplying both sides on the left by X^*W gives

$$X^*W\left(\frac{1}{N_{\text{blk}}}XX^*W\hat{\Phi}\right) = X^*W\hat{\Phi}\hat{\Lambda}, \tag{3}$$

and using associativity,

$$\frac{1}{N_{\text{blk}}}\left(X^*WX\right)\left(X^*W\hat{\Phi}\right) = \left(X^*W\hat{\Phi}\right)\hat{\Lambda}. \tag{4}$$

Defining the reduced correlation matrix $R \equiv \frac{1}{N_{\text{blk}}}X^*WX$ and the reduced eigenvectors $\Psi \equiv X^*W\hat{\Phi}$, this becomes

$$R\Psi = \Psi\hat{\Lambda}. \tag{5}$$

The reduced system is computationally tractable because R is only $N_{\text{blk}} \times N_{\text{blk}}$. The physical influence of the grid weights W is embedded inside R. Solving Equation (5) yields the modal energies $\hat{\Lambda}$ and the orthonormal reduced eigenvectors Ψ (satisfying $\Psi^*\Psi = I$).

Spatial mode reconstruction

Because X^*W is rectangular, the definition $\Psi = X^*W\hat{\Phi}$ cannot be inverted directly to recover the spatial modes. Following Sirovich's snapshot-method assumption[8], the spatial modes are constructed as a linear combination of the original frequency snapshots, weighted by the reduced eigenvectors:

$$\hat{\Phi}_{\text{raw}} = X\Psi. \tag{6}$$

Orthonormality scaling

The final spatial modes must satisfy the weighted discrete orthonormality condition $\hat{\Phi}^*W\hat{\Phi} = I$. Testing the raw modes,

$$\hat{\Phi}^*_{\text{raw}}W\hat{\Phi}_{\text{raw}} = (X\Psi)^*W(X\Psi) = \Psi^*(X^*WX)\Psi. \tag{7}$$

From Equation (5), $X^*WX\,\Psi = N_{\text{blk}}\Psi\hat{\Lambda}$, giving

$$\hat{\Phi}^*_{\text{raw}}W\hat{\Phi}_{\text{raw}} = N_{\text{blk}}\Psi^*\Psi\hat{\Lambda} = N_{\text{blk}}\hat{\Lambda}. \tag{8}$$

The squared length of the raw modes is therefore scaled by $N_{\text{blk}}\hat{\Lambda}$. Because the orthonormality condition is quadratic in $\hat{\Phi}$, the correctly scaled SPOD modes are

$$\hat{\Phi} = \frac{1}{\sqrt{N_{\text{blk}}\hat{\Lambda}}}\,X\Psi. \tag{9}$$

Equation (5) delivers the modal energies $\hat{\Lambda}$, and Equation (9) delivers the spatial modes $\hat{\Phi}$.

2.2 Procedure Summary

The complete SPOD procedure derived above is summarised schematically in Figure 1.

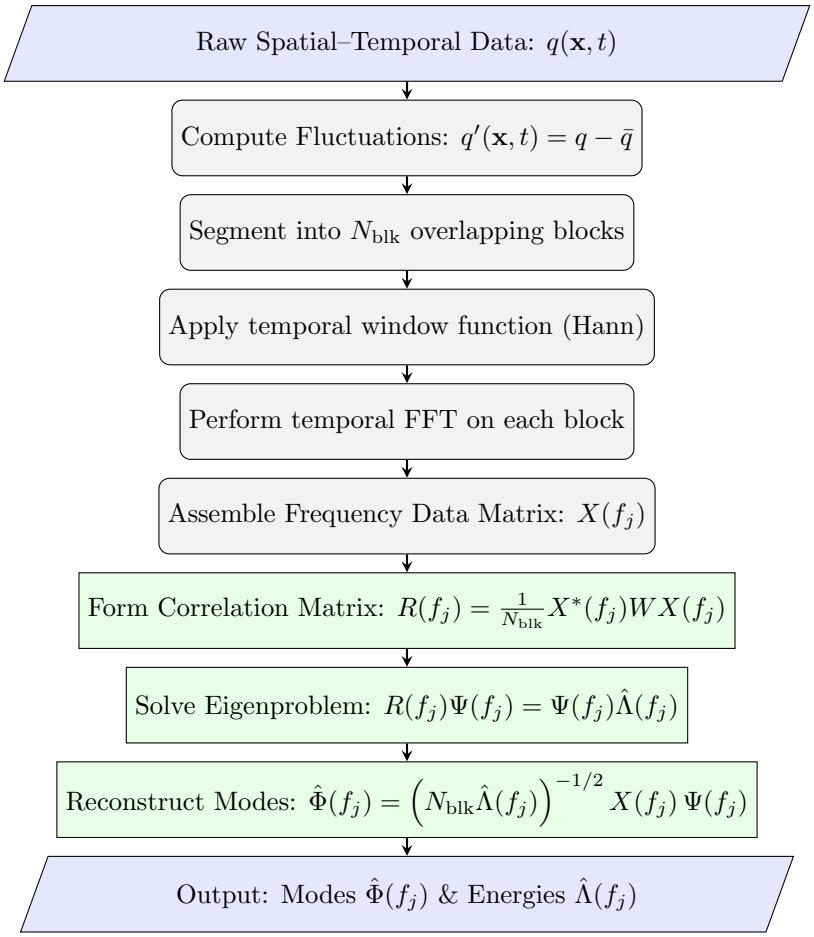

Figure 1: SPOD procedure flowchart. Blue trapezoids denote data inputs and outputs; grey blocks denote preprocessing steps; green blocks denote the core mathematical operations of the weighted Method of Snapshots.

3 Results

The pipeline produced 26 frequency bins from $N_{\text{blk}} = 19$ overlapping blocks, giving $\Delta f = 2.667$ and a Nyquist frequency of $f_{\text{Ny}} = 66.67$. What follows: first the eigenvalue spectrum, then mode-dominance diagnostics, then the spatial structure of the leading modes, and finally their component-energy split.

3.1 Reconstruction Fidelity and Modal Convergence

To verify the mathematical consistency of the SPOD basis and the validity of the energy truncation threshold, a single Welch block at $f = 2.67$ was projected onto the leading K_{90} spatial modes, where K_{90} is defined as the minimum number of modes whose cumulative eigenvalue fraction reaches 90%:

$$K_{90} = \min\left\{ K : \frac{\sum_{k=1}^{K} \hat{\Lambda}_k}{\sum_{k=1}^{N_{\text{blk}}} \hat{\Lambda}_k} \geq 0.90 \right\}. \tag{10}$$

At $f = 2.67$ this yields $K_{90} = 10$ out of 19 available modes, with a cumulative eigenvalue fraction of 0.9095.

The reconstruction is performed via a W-weighted Galerkin projection. The modal amplitude vector $\boldsymbol{a} \in \mathbb{C}^{K_{90}}$ is computed by projecting the original frequency-domain snapshot $\boldsymbol{q}_{\text{orig}}$ onto each retained mode through the grid-weighted inner product:

$$a_k = \hat{\boldsymbol{\phi}}_k^* \, W \, \boldsymbol{q}_{\text{orig}}, \qquad k = 1, \ldots, K_{90}, \tag{11}$$

where W is the diagonal spatial weight matrix encoding the non-uniform grid volumes. Each scalar a_k is therefore the amplitude and phase by which mode k must be weighted to best represent this particular snapshot in the W-inner-product sense. The reconstructed field and residual are then:

$$\boldsymbol{q}_{\text{rec}} = \sum_{k=1}^{K_{90}} a_k \, \hat{\boldsymbol{\phi}}_k, \qquad \boldsymbol{q}_{\text{res}} = \boldsymbol{q}_{\text{orig}} - \boldsymbol{q}_{\text{rec}}. \tag{12}$$

The reconstruction fidelity is quantified by the W-weighted energy ratio:

$$\frac{E_{\text{rec}}}{E_{\text{orig}}} = \frac{\|\boldsymbol{q}_{\text{rec}}\|_W^2}{\|\boldsymbol{q}_{\text{orig}}\|_W^2} = 0.9205, \qquad \frac{E_{\text{res}}}{E_{\text{orig}}} = 0.0795, \tag{13}$$

where $\|\boldsymbol{v}\|_W^2 = \boldsymbol{v}^* W \boldsymbol{v}$. The realised ratio of 92.05% slightly exceeds the theoretical eigenvalue cumulative fraction of 90.95%, consistent with expected block-to-block statistical fluctuation around the ensemble-averaged threshold.

Figure 2 shows the original DNS data, reconstructed coherent field, and residual for all three velocity components at mid-span. The residual panels are visually devoid of organised spatial structure, confirming that the K_{90} truncation retains all coherent content and discards only incoherent background fluctuations.

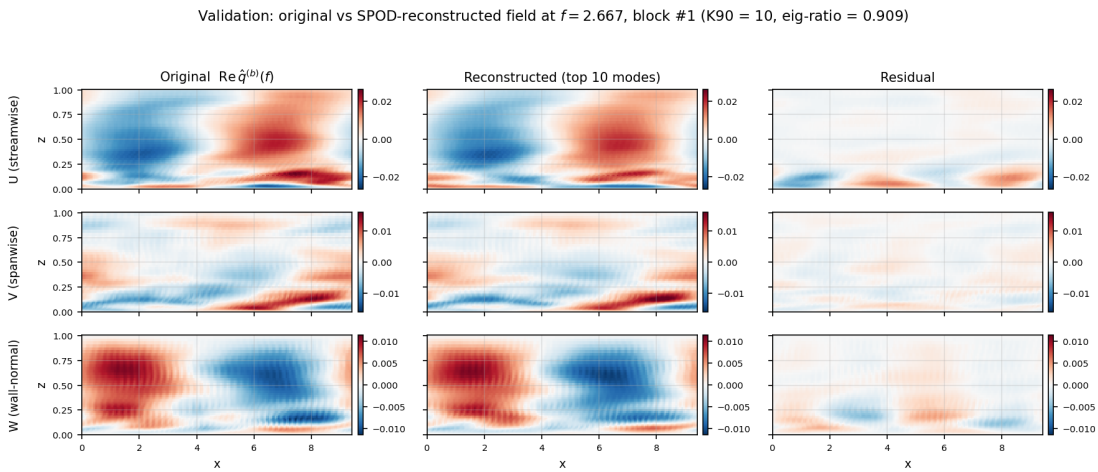

Figure 2: Modal reconstruction validation at $f = 2.67$, Welch block 1. Columns: (left) original DNS snapshot $\mathrm{Re}\,\hat{\boldsymbol{q}}^{(b)}(f)$, (centre) reconstructed field using the leading $K_{90} = 10$ SPOD modes, (right) residual $\boldsymbol{q}_{\text{res}}$. Rows show the streamwise (U), spanwise (V), and wall-normal (W) velocity components; within each row all three columns share the same colour scale. The reconstruction recovers 92.05% of the block energy ($E_{\text{res}}/E_{\text{orig}} = 0.0795$), in close agreement with the theoretical eigenvalue cumulative fraction of 90.95%.

3.2 SPOD Eigenvalue Spectrum

Figure 3 presents the complete SPOD eigenvalue spectrum across all 26 resolved frequencies, displayed on both linear-frequency/log-energy and log–log axes.

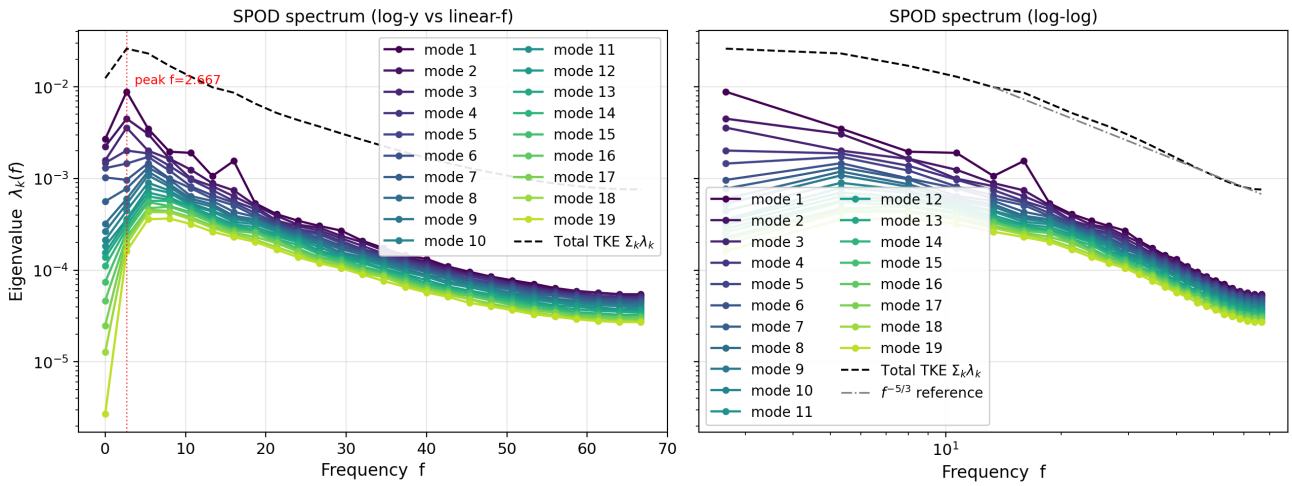

Figure 3: SPOD eigenvalue spectrum across all resolved frequencies. Each curve corresponds to a single mode rank (modes 1–19, darkest to lightest); the dashed black line is the total energy per frequency $\sum_k \hat{\Lambda}_k$. Left: log-energy versus linear frequency, with the global peak at $f = 2.67$ (red dotted line). Right: log–log axes, with a dash-dotted $f^{-5/3}$ reference line.

Three frequency bands stand out. Below $f \approx 5.33$ the leading mode is well clear of the others, with a global peak at $f = 2.67$ where $\hat{\Lambda}_1 = 8.84 \times 10^{-3}$ and the coherence ratio $\hat{\Lambda}_1 / \sum_k \hat{\Lambda}_k = 0.339$. This is a single coherent

structure sitting at one frequency. Between $f \approx 5.33$ and $f \approx 16.0$ the leading mode still keeps a comfortable lead. Past $f \approx 18.67$ the curves bunch together and the spectrum flattens out into a broadband, incoherent turbulent regime where all 19 ranks carry roughly the same energy.

A small bump near $f = 16.0$ shows up in mode 1 and mode 2, breaking the otherwise smooth decay of the leading-mode curve. This bump has the largest rank separation in the whole spectrum, $\hat{\Lambda}_1/\hat{\Lambda}_2 = 2.09$ (Section 3.3), so $f = 16.0$ is included as a target frequency in Section 3.4.

On the log-log axes, the total energy curve $\sum_k \hat{\Lambda}_k$ follows the Kolmogorov $f^{-5/3}$ slope reasonably well between $f \approx 6$ and the Nyquist frequency $f = 66.67$, suggesting the small-scale fluctuations are adequately resolved.

3.3 Spectral Energy Distribution and Mode Dominance

Three ratios are useful for picking out the dominant modes: the leading-mode energy $\hat{\Lambda}_1$, the coherence ratio $\hat{\Lambda}_1/\sum_k \hat{\Lambda}_k$, and the rank separation $\hat{\Lambda}_1/\hat{\Lambda}_2$. Across the whole spectrum the total modal energy comes to $\sum_f \sum_k \hat{\Lambda}_k(f) = 1.51 \times 10^{-1}$. Table 1 shows the global peak sitting at $f = 2.67$.

Table 1: Top five frequencies ranked by leading mode energy $\hat{\Lambda}_1$. Columns: $\sum \hat{\Lambda}_k$ is the total energy at that frequency; %Total is its share of the spectrum-wide total; $\hat{\Lambda}_1$ is the energy in mode 1; $\hat{\Lambda}_1/\sum \hat{\Lambda}_k$ is mode 1's share of the local energy; $\hat{\Lambda}_1/\hat{\Lambda}_2$ is the rank-1/2 separation; the last column is the number of modes needed to capture 90% of the local energy. The $f = 0$ entry reflects residual block-to-block variation in the mean velocity field rather than a true zero-frequency mode, since each block undergoes independent mean subtraction prior to the FFT.

Rank	f	$\sum \hat{\Lambda}_k$	**%Total**	$\hat{\Lambda}_1$	$\hat{\Lambda}_1/\sum \hat{\Lambda}_k$	$\hat{\Lambda}_1/\hat{\Lambda}_2$	**Modes** $\geq 90\%$
1	2.6667	2.606e-02	17.23	8.835e-03	0.339	1.97	10
2	5.3333	2.309e-02	15.27	3.493e-03	0.151	1.14	14
3	0.0000	1.241e-02	8.21	2.692e-03	0.217	1.21	8
4	8.0000	1.692e-02	11.19	1.958e-03	0.116	1.19	16
5	10.667	1.289e-02	8.53	1.902e-03	0.148	1.53	16

At the peak frequency $f = 2.67$ the leading mode picks up 33.9% of the local energy, and the rank separation $\hat{\Lambda}_1/\hat{\Lambda}_2 = 1.97$ sits just under the 2.0 threshold typically used for rank-1 dominance. The largest separation in the spectrum is at $f = 16.0$ ($\hat{\Lambda}_1/\hat{\Lambda}_2 = 2.09$), so even though that frequency carries less absolute energy, the structure there is unusually distinct from the rest. These two, together with $f = 8.0$ (broader coherent regime) and $f = 53.33$ (turbulent reference) gives four target frequencies for the rest of the analysis. Together, the top three frequencies in the table account for over 40% of the total spectral energy.

The coherence ratio also pinpoints where the flow loses coherence. With $N_{\text{blk}} = 19$ the incoherent baseline is $1/N_{\text{blk}} = 0.053$ and the conventional threshold is $2/N_{\text{blk}} = 0.105$. Mode 1 exceeds 0.105 for all $f \leq 16.0$; the first frequency where it falls below this threshold is $f = 18.67$. Past that point, the energy spreads almost evenly over the blocks, and the number of modes needed for 90% of the variance saturates at 16–17 out of a maximum of 19. The upper edge of the coherent regime is therefore around $f \approx 18.67$.

3.4 Vertical Structure of the Three Leading Modes Across Frequencies

Figures 4, 5, and 6 show the phase-aligned real parts of the first three SPOD modes at four target frequencies ($f = 2.67$, 8.0, 16.0, and 53.33), sliced at mid-span ($y = 4.737$). The four frequencies cover the three regimes identified in Section 3.2: the primary coherent peak ($f = 2.67$), the intermediate range ($f = 8.0$ and $f = 16.0$), and the incoherent reference ($f = 53.33$).

Canonical phase alignment

Since SPOD modes are complex, each one must be rotated to a fixed phase before plotting; otherwise the snapshot picked at random will look different for every mode and frequency. The real-part energy fraction (i.e. the share of the total mode energy carried by the real part after rotation) gives a useful indicator of wave type. If almost all the energy sits in the real part, the mode is essentially a standing wave. If the split is close to 50/50, it is a travelling wave: the imaginary part carries the same amount of energy as the real part, which is what one expects for a structure that propagates as $e^{i(kx-\omega t)}$. Anything in between is a mixture.

To visualise each mode at the same canonical phase, we apply a global phase rotation $e^{-i\varphi^\star}$ chosen to maximise the real-part energy:

$$\varphi^\star = \tfrac{1}{2} \arg\left(\sum_n W_n \, \hat{\Phi}_n^2 \right), \tag{14}$$

where W_n are the spatial weights from Section 4.2.

For all twelve mode–frequency combinations the real-part fraction sits at about 50% (within 0.3%). Every structure shown is a downstream-propagating wave.

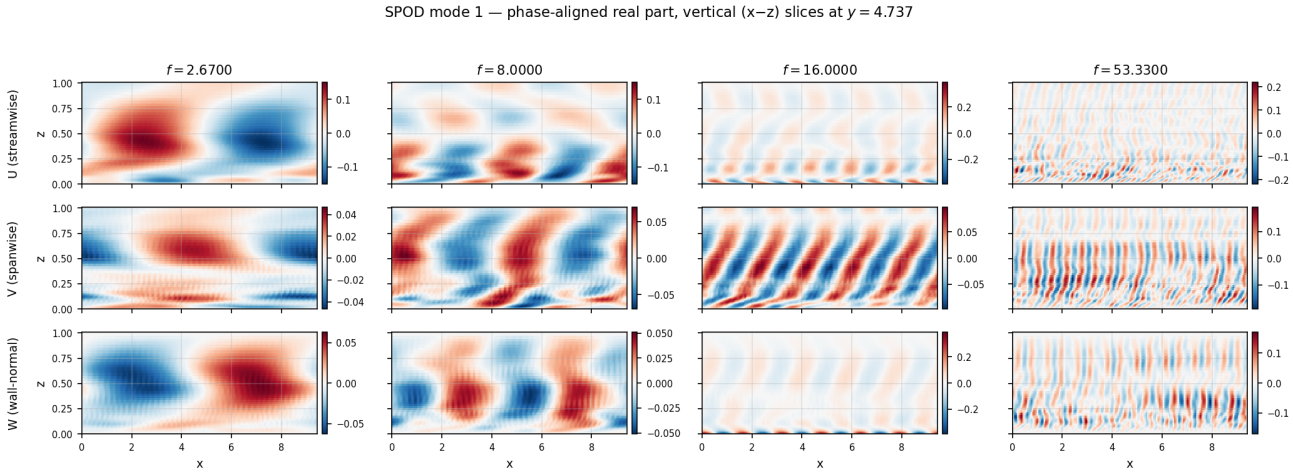

Figure 4: Phase-aligned real part of SPOD mode 1 at four frequencies (f = 2.67, 8.0, 16.0, 53.33), plotted as vertical (x–z) slices at mid-span (y = 4.737). Rows: U, V, W velocity components. Columns: increasing frequency from left to right. Each panel is normalised to its own maximum amplitude.

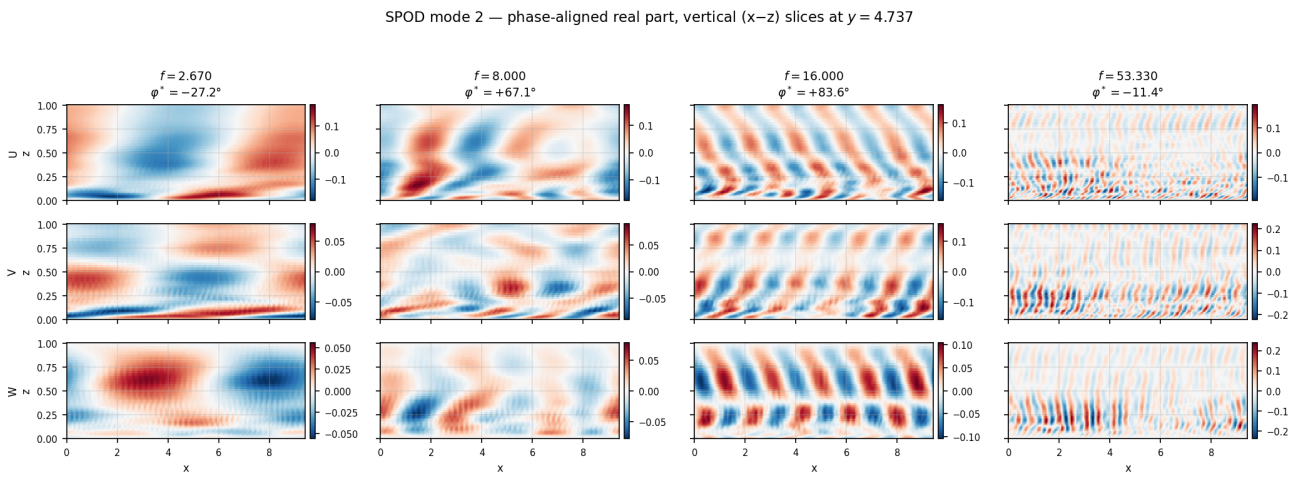

Figure 5: Phase-aligned real part of SPOD mode 2 at four frequencies, plotted as vertical (x–z) slices at mid-span. Layout identical to Figure 4.

Mode 1: dominant coherent structure

Mode 1 changes character with frequency (Figure 4). At $f = 2.67$ it appears as a large, smooth structure that spans the full streamwise extent of the slice; the partition is strongly streamwise ($U : V : W = 68.0 : 21.6 : 10.4$) and the energy sits in the lower-to-mid domain ($z_{\text{peak}} = 0.39$, $\bar{z} = 0.49$). At $f = 8.0$ the structure has contracted and tilted forward, as expected from mean-shear deformation. At $f = 16.0$ the behaviour changes qualitatively: W rises to 35.5%, approaching parity with U (44.1%) — the only frequency in the spectrum where this occurs — and the mode collapses onto a thin near-surface layer. At $f = 53.33$ all structure is lost and the partition approaches near-isotropy ($U : V : W \approx 26\% : 41\% : 33\%$).

Modes 2 and 3: independent sub-dominant structures

At $f = 2.67$ modes 1, 2, and 3 together carry about 65% of the spectral energy. Modes 2 and 3 are not just shifted copies of mode 1: mode 2 lives in the upper half of the domain ($z \gtrsim 0.5$) while mode 3 is squeezed into the lower quarter ($z \lesssim 0.25$). Their component splits ($U : V : W = 72.2 : 17.7 : 10.1$ and $76.0 : 14.6 : 9.5$) are more U-heavy than mode 1, and they have different streamwise wavelengths too. So the three modes are genuinely

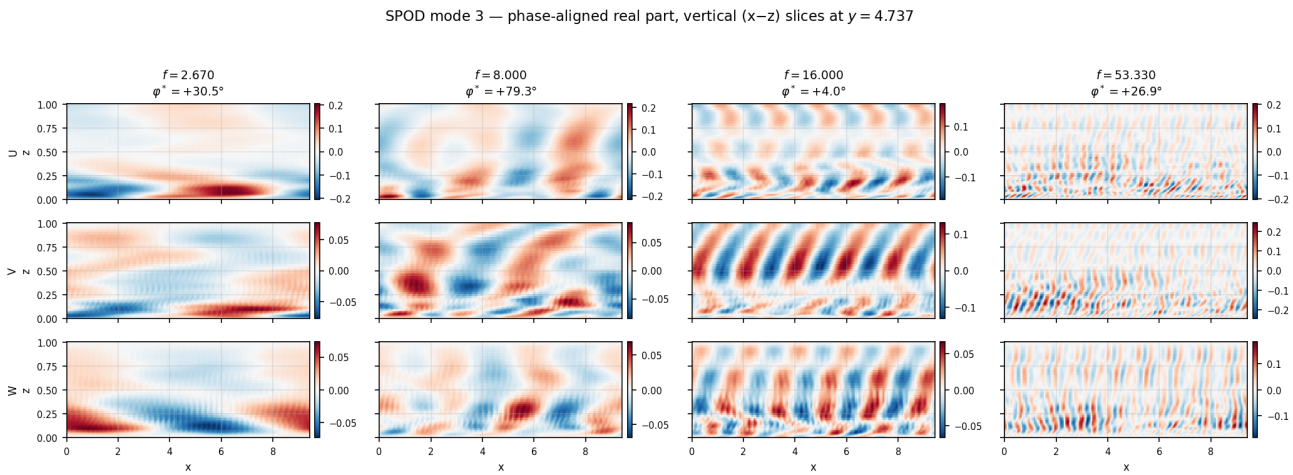

Figure 6: Phase-aligned real part of SPOD mode 3 at four frequencies, plotted as vertical (x–z) slices at mid-span. Layout identical to Figure 4.

independent structures coexisting at the same frequency, not harmonics or phase-shifted versions of one another. By $f = 53.33$ they have all collapsed onto roughly the same incoherent split ($U : V : W \approx 26$–$27 : 40$–$41 : 33$).

Component-energy cascade across rank and frequency

A clean trend runs across the three modes. In the coherent regime the U share *increases* with rank ($68.0 \rightarrow 72.2 \rightarrow 76.0$ at $f = 2.67$), so higher-rank modes look more streak-like. Going up in frequency, the U share *drops* for every mode and the split moves towards isotropy. At $f = 53.33$ all three modes share the same faintly V-biased split. The wave-driven shear leaves its streamwise-anisotropy fingerprint most clearly on mode 1 at low frequency, and that fingerprint fades both with higher rank and with higher frequency.

3.5 Wall-Normal Energy Distribution and Component-Energy Cascade

Two further diagnostics for mode 1 round out the picture from Section 3.4. Panel (a) of Figure 7 presents the wall-normal energy distribution $E(z)$ at the four target frequencies, and panel (b) presents the U, V, W component-energy shares across all 26 resolved frequencies.

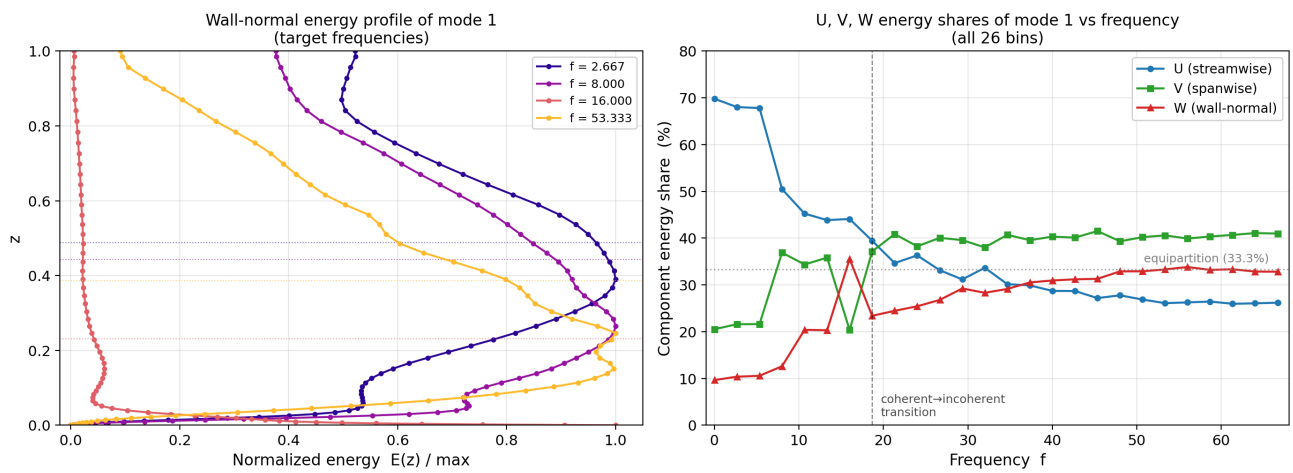

Figure 7: (a) Wall-normal energy profile of SPOD mode 1 at the four target frequencies, normalised to the profile peak. Dotted horizontal lines mark the energy centroids $\bar{z}$. (b) U, V, and W component-energy shares of mode 1 across all 26 frequency bins. The grey dotted line marks equipartition (1/3) and the vertical dashed line marks the coherent–incoherent transition.

Wall-normal energy profiles

The four profiles are clearly different from one another; each target frequency picks up a different wall-normal arrangement. The key diagnostics are summarised in Table 2.

Table 2: Wall-normal localisation of SPOD mode 1 at the four target frequencies. z_{peak} is the height of the energy maximum and $\bar{z}$ is the volume-weighted centroid.

f	z_{peak}	$\bar{z}$
2.67	0.390	0.488
8.0	0.265	0.443
16.0	0.000	0.231
53.33	0.247	0.386

The trend reflects a well-known property of wall-bounded turbulence: large-scale, low-frequency structures organise in the bulk of the domain, while high-frequency fluctuations are increasingly confined to the near-wall region where velocity gradients are steepest and small-scale turbulence is generated. The SPOD wall-normal profiles recover this hierarchy directly from the data. The primary coherent mode at $f = 2.67$ peaks at $z_{\text{peak}} = 0.39$ well into the interior ($\bar{z} = 0.49$), consistent with a wave-coherent structure whose length scale is comparable to the domain height. As frequency increases through $f = 8.0$ ($z_{\text{peak}} = 0.27$, $\bar{z} = 0.44$) the energy centroid descends progressively toward the wall. At $f = 16.0$ the mode is pinned entirely to the surface ($z_{\text{peak}} = 0.000$, $\bar{z} = 0.23$): this is the only frequency where the dominant coherent structure is a near-wall phenomenon, consistent with its identification in Section 3.4 as the frequency of maximum rank separation and peak W energy. The incoherent reference at $f = 53.33$ breaks from this monotonic trend with two peaks: a main one near $z = 0.25$ and a secondary one at $z = 0.95$, which is the wall-normal signature of coherence breakdown: energy is no longer organised into a single connected structure but is scattered across the domain.

Component-energy cascade across frequency

Panel (b) shows the anisotropy cascade in mode 1. In the primary coherent regime ($f \lesssim 5.33$) the mode is heavily U-dominated ($U : V : W \approx 68 : 21.6 : 10$), reflecting the streamwise organisation of the wave-forced structures. Through the intermediate range the U share drops sharply and W jumps up, hitting 35.5% at $f = 16.0$: the only frequency at which W overtakes V. This is consistent with the near-wall localisation at that frequency — wall-normal velocity fluctuations are enhanced relative to the streamwise component in the viscous and buffer layers. Past the coherent–incoherent transition at $f \approx 18.67$ the three shares flatten out: U crosses equipartition at $f = 26.67$ and the split settles at roughly $27 : 40 : 33$. The small-scale fluctuations are close to isotropic, with a slight V bias.

4 How the code works?

This section walks through the actual implementation, with an emphasis on the choices that were made for memory efficiency. The pipeline has four phases: data ingestion and matrix setup, preprocessing (windowing and fluctuation extraction), spectral estimation, and mode reconstruction.

4.1 Data Ingestion and Matrix Setup

The DNS writes its output through Fortran's unformatted stream access (`ACCESS='STREAM'`), so on disk it is just a long run of raw bytes with no record markers. SPOD needs fast, non-sequential access to time blocks, so the bytes have to be reorganised first. The reader walks through the stream and pulls out exactly $N_x \times N_y \times N_z$ values for each of the three velocity components at each timestep. These get stacked into a 2D snapshot matrix Q where rows are flattened spatial coordinates and columns are time snapshots. The full Q is too big for RAM, so it is written to disk as a memory-mapped array via the `build_Q_memmap` routine. Any time block is then one disk read away, without ever loading the whole matrix.

4.2 Spatial Weight Matrix Computation

The weight matrix W is what makes Equation (1) well-defined: it is the discrete quadrature rule for the inner product. Each grid point is weighted by the volume it represents.

For the present computational domain, the horizontal grid spacing is uniform ($\Delta x = L_x/N_x$ and $\Delta y = L_y/N_y$), while the vertical coordinates z_k are strictly increasing but non-uniform. The discrete vertical weight at index k is computed using the trapezoidal rule:

$$\Delta z_k = \tfrac{1}{2}(z_{k+1} - z_{k-1}), \tag{15}$$

with one-sided differences applied at the domain boundaries. The local spatial weight for any node is the three-dimensional control volume $\Delta V_{ijk} = \Delta x\, \Delta y\, \Delta z_k$.

A simple sanity check is to sum all the weights and compare with the domain volume:

$$\sum_{k=0}^{N_z-1}\sum_{j=0}^{N_y-1}\sum_{i=0}^{N_x-1}\Delta x\,\Delta y\,\Delta z_k = L_x L_y\,(z_{\max}-z_{\min}). \tag{16}$$

For the present grid, the computed weight sum matches the analytical value $9\pi^2 \approx 88.83$ to machine precision.

The state vector stacks three velocity components, all carrying the same dimensional units, so no extra component-wise scaling is needed. The full W is just the per-cell volumes tiled three times into a 1D diagonal array.

4.3 Block Segmentation and Fluctuation Extraction

To estimate a robust CSD, the snapshot matrix Q is partitioned into an ensemble of shorter temporal blocks. If the blocks were extracted sequentially without overlap (as in the Bartlett method[5]), the abrupt truncation at the block boundaries would introduce artificial high-frequency noise into the Fourier transform - spectral leakage. To mitigate this, a Hann window[6] is applied to each block, smoothly tapering the velocity signal to zero at the beginning and end of the segment and enforcing strict periodicity.

The Hann window introduces a secondary complication: by forcing the boundaries to zero, it discards physical content near the edges of every block. To compensate, the algorithm uses Welch's method with 50% block overlap, ensuring that turbulence attenuated at the edge of one block sits at the unattenuated peak of an adjacent one. The overlap also increases the number of available block realisations, raising the statistical degrees of freedom and reducing the variance of the spectral estimate. Because true turbulent structures are coherent across time, ensemble-averaging across many overlapping blocks suppresses random noise while preserving distinct energy peaks of the underlying physical phenomena.

The reader pulls each block's columns straight from the memmap, one block at a time. Once a block is in memory, its time-mean is subtracted row by row, leaving the fluctuation

$$q'(\mathbf{x},t) = q(\mathbf{x},t) - \bar{q}_{\text{block}}(\mathbf{x}). \tag{17}$$

The mean is subtracted here rather than upstream in the Fortran solver. Keeping the full instantaneous velocity in the binary output makes the dataset reusable for other analyses without rerunning the DNS, and isolating fluctuations is something the post-processing pipeline can handle anyway.

If the mean were left in, its energy would swamp everything else and show up as an enormous spike at $f = 0$, drowning out the turbulence SPOD is supposed to be looking at. Removing the block-wise mean focuses the decomposition on fluctuations about each block's local baseline. Some residual energy at $f = 0$ remains and is visible in the spectrum: this is just the block-to-block drift in the mean flow, which is unavoidable when the underlying state is not strictly stationary.

4.4 Spectral Transformation and Frequency Matrix Assembly

A real FFT is applied along the time axis of each block. The resulting Fourier coefficients are then rescaled by the factor

$$\kappa = \sqrt{\frac{\Delta t}{\sum_{n=1}^{N_t} w_n^2}}, \tag{18}$$

where w_n is the Hann window amplitude at sample n, Δt is the snapshot time spacing, and $N_t = 50$ is the block length (number of snapshots per block, not the number of blocks $N_{\text{blk}} = 19$). This scaling compensates for the variance attenuation introduced by the windowing and ensures that the final eigenvalues $\hat{\Lambda}(f)$ represent power spectral density with correct physical dimensions.

Textbook SPOD treats the FFT output as a single $N_{\text{space}} \times N_{\text{blk}} \times N_f$ tensor, but that does not fit in memory for a DNS this size. Instead, the code processes one frequency at a time. For a chosen f_j, it pulls the matching Fourier coefficient from every block, stacks them into $X(f_j) \in \mathbb{C}^{N_{\text{space}} \times N_{\text{blk}}}$, and runs the eigenvalue problem on that. Everything needed at f_j is in this one matrix.

4.5 Assembly of the Reduced Correlation Matrix

Building R in one shot would mean forming the full X^*WX product over the whole spatial domain, which is again too large. But matrix multiplication is distributive over partitioned rows: the spatial domain can be split into chunks (here, 20,000 grid points each), each chunk's contribution to R computed separately, and the results summed. After dividing by N_{blk} at the end, this gives the exact R without ever holding the full $X(f_j)$ in memory.

R should be Hermitian by construction, but floating-point roundoff from the chunked sum breaks this very slightly. To keep the eigenvalue solver well-behaved, the final matrix is symmetrised explicitly: $R \leftarrow \frac{1}{2}(R + R^*)$ before decomposition.

4.6 Reduced Eigenvalue Decomposition

Once $R(f_j)$ is built and symmetrised, a standard Hermitian eigensolver gives the modal energies $\hat{\Lambda}$ and the reduced eigenvectors Ψ. The eigenpairs are then sorted by energy in descending order, so $\hat{\Phi}_1$ is the most energetic mode at f_j by construction.

Kinetic energy must be non-negative, but floating-point error in building R can leave a few eigenvalues at $\sim -10^{-16}$. These are clipped to zero before the spatial reconstruction step.

4.7 Spatial Mode Reconstruction and Verification

The 3D spatial modes follow from Equation (9). As with R, computing $\hat{\Phi} = (N_{\text{blk}}\hat{\Lambda})^{-1/2} X\Psi$ in one go is too large, so the code reads $X(f_j)$ in spatial chunks, projects each chunk onto $(N_{\text{blk}}\hat{\Lambda})^{-1/2}\Psi$, and writes the result directly to disk.

As a check, $\hat{\Phi}^* W \hat{\Phi}$ is computed in chunks and compared to the identity matrix. The maximum residual is $\sim 10^{-15}$, so the modes are orthonormal in the weighted inner product to machine precision.

4.8 Eigenvalues as Power Spectral Density

Each $\hat{\Lambda}_k(f_j)$ is the energy carried by the k-th mode at frequency f_j. Because the Fourier coefficients were rescaled by κ (Equation (18)), the eigenvalues come out with units of power spectral density: energy per unit frequency.

The sum $\sum_{k=1}^{N_{\text{blk}}} \hat{\Lambda}_k(f_j)$ yields the total kinetic energy density at frequency f_j. Integrating across all frequencies recovers the total turbulent kinetic energy of the flow:

$$\text{TKE} = \sum_{f_j} \sum_{k=1}^{N_{\text{blk}}} \hat{\Lambda}_k(f_j)\, \Delta f, \tag{19}$$

where Δf is the frequency resolution. For the present dataset this gives $\text{TKE} = 1.512 \times 10^{-1} \times 2.667 = 4.03 \times 10^{-1}$. The full SPOD spectrum $\hat{\Lambda}_k(f_j)$ thus describes how energy is split across spatial scales (the rank k) and temporal scales (the frequency f). The single largest eigenvalue points to the most energetic coherent structure in the flow.

4.9 Full Spectrum Execution

With the single-frequency pipeline working, the code loops over all the FFT frequencies; the physical frequency axis is set by the snapshot sampling rate Δt.

A few practical points keep the loop stable. At the start of each iteration the memory-mapped file handles are released and Python's garbage collector is run, which avoids both memory blow-up and operating-system file-lock errors during long runs. The orthonormality check $\hat{\Phi}^* W \hat{\Phi} = I$ is only run on the first frequency, since the same projection structure is reused for every frequency afterwards. Inside the loop, the 3D spatial modes go to disk directly and the eigenvalues $\hat{\Lambda}$ are collected into a single global array. That array is the SPOD spectrum used in Section 3.

A Complete SPOD Eigenvalue Spectrum

Table 3: Full SPOD spectral data across all 26 resolved frequencies ($N_{\text{blk}} = 19$ blocks). The %Total column gives each frequency's share of the total kinetic energy.

idx	f	$\sum \hat{\Lambda}_k$	**%Total**	$\hat{\Lambda}_1$	$\hat{\Lambda}_1/\sum \hat{\Lambda}_k$	$\hat{\Lambda}_1/\hat{\Lambda}_2$	**Modes** $\geq 90\%$
0	0.00	1.2409e-02	8.21	2.6916e-03	0.217	1.21	8
1	2.67	2.6058e-02	17.23	8.8353e-03	0.339	1.97	10
2	5.33	2.3090e-02	15.27	3.4933e-03	0.151	1.14	14
3	8.00	1.6923e-02	11.19	1.9583e-03	0.116	1.19	16
4	10.67	1.2894e-02	8.53	1.9019e-03	0.148	1.53	16
5	13.33	9.8843e-03	6.54	1.0629e-03	0.108	1.20	16
6	16.00	8.6232e-03	5.70	1.5526e-03	0.180	2.09	16
7	18.67	6.5599e-03	4.34	5.3657e-04	0.082	1.05	16
8	21.33	5.1826e-03	3.43	4.0844e-04	0.079	1.05	17
9	24.00	4.3324e-03	2.87	3.4575e-04	0.080	1.10	17
10	26.67	3.6861e-03	2.44	3.0333e-04	0.082	1.05	17
11	29.33	3.1061e-03	2.05	2.7042e-04	0.087	1.14	17
12	32.00	2.6392e-03	1.75	2.0960e-04	0.079	1.04	17
13	34.67	2.2513e-03	1.49	1.7357e-04	0.077	1.06	17
14	37.33	1.9313e-03	1.28	1.4564e-04	0.075	1.05	17
15	40.00	1.6804e-03	1.11	1.3238e-04	0.079	1.07	17
16	42.67	1.4699e-03	0.97	1.0954e-04	0.075	1.03	17
17	45.33	1.2952e-03	0.86	9.6010e-05	0.074	1.05	17
18	48.00	1.1652e-03	0.77	8.5129e-05	0.073	1.03	17
19	50.67	1.0495e-03	0.69	7.7515e-05	0.074	1.05	17
20	53.33	9.5332e-04	0.63	7.0977e-05	0.074	1.05	17
21	56.00	8.7816e-04	0.58	6.3759e-05	0.073	1.04	17
22	58.67	8.2588e-04	0.55	5.9304e-05	0.072	1.02	17
23	61.33	7.8814e-04	0.52	5.6850e-05	0.072	1.01	17
24	64.00	7.6450e-04	0.51	5.4697e-05	0.072	1.01	17
25	66.67	7.5853e-04	0.50	5.4538e-05	0.072	1.02	17
Total	—	1.5120e-01	100.00%	—	—	—	—

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